Raghunandan Rajkumar

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Portfolio: https://rose-ember-forge.github.io/

Physicist turned data scientist.
Passionate about turning complex datasets into actionable insight
through rigorous analysis and machine learning.

Experience

Research Scholar Jul 2018 – Nov 2021
Department of Physics, Pondicherry University *Pondicherry, India*

- Independently led a three-year research project using artificial neural networks and deep learning to model the functioning of biological neural networks.
- As a part of a research team, analyzed various EEG datasets and used Entropy measures to recognise human emotions, resulting in a published paper.
- **Publication:** *EEG-based Human Emotion Recognition using Entropy as a Feature Extraction Measure* `https://doi.org/10.1186/s40708-021-00141-5`.

Workshop Organizer 2019
Pondicherry University *Pondicherry, India*

- Conceived, planned, and led a technical workshop on LaTeX for physics students, managing everything from curriculum design to on-the-day facilitation.
- Translated complex typesetting concepts into accessible, hands-on content-making technical tools approachable for a non-specialist audience.
- Successfully promoted department-wide use of LaTeX as a better, streamlined method of academic typesetting.

Teacher 2017
Yuvaraja's College *Mysuru, India*

- Delivered master's-level physics courses, adapting content to diverse learning styles and fostering an environment where students felt comfortable engaging with challenging material.
- Mentored graduate students through complex problem-solving, building their confidence alongside their technical understanding.
- Delivered master's-level lectures in C-programming, connecting computational thinking to students' existing physics background to make abstract concepts concrete.

Technical Skills

Languages: Python, R, SQL, LaTeX, C
Data Science & Analysis: Data Visualization, NLP, Sentiment Analysis, Machine Learning, Deep Learning, Artificial Neural Networks
Tools & Libraries: Pandas, Scikit-Learn, PowerBI, Folium, Plotly, Spreadsheet software, Regular Expressions (RegEx)
Soft Skills: Problem Solving, Communication, Time Management, Leadership, Organization

Education

Master of Science in AI for Health Stockholm, Sweden
Stockholm University *2024 – 2026*

Master of Science (Physics) – CGPA: 6.86/10 Mysuru, India
Yuvaraja's College, University of Mysore *2016 – 2018*

Bachelor of Science (Physics, Chemistry, Mathematics) – 68.9% Mysuru, India
Yuvaraja's College, University of Mysore *2013 – 2016*

Projects

SpaceX Launch Analysis (Data Science Capstone) | *SQL, Python, Folium, Plotly*

- Independently collected, cleaned, and engineered features from SpaceX launch data via REST API and Wikipedia web scraping, preparing the dataset for classification tasks.
- Conducted Exploratory Data Analysis using SQL queries and interactive visualizations (Folium, Plotly), uncovering patterns in launch success rates across sites and payload types.
- Trained and evaluated four classifiers (SVM, KNN, Logistic Regression, Decision Tree), identifying Decision Tree as the top performer.

Book Recommendation Web App | *Python, NLP, TF-IDF, Random Forest, Flask*

- Collaboratively designed and built a full-stack book recommendation web application in Flask, integrating NLP-based ML models as modular backend components.
- Developed a sentiment analysis pipeline using TF-IDF vectorization and tree-based models (Decision Tree, Random Forest) to classify review polarity.
- Built a genre prediction module to classify books by summary using text classification techniques; evaluated performance using AUC and accuracy metrics.
- Handled end-to-end data preprocessing of raw review data for robust feature extraction, from cleaning through to model-ready feature matrices.

PINN-based modeling of Greenland Ice Sheet melt (Master's thesis project) | *Deep Learning, ANNs*

- Developed, as part of a two-person research team, a Physics-Informed Neural Network (PINN) model in PyTorch to simulate the thermo-mechanical dynamics of the Greenland Ice Sheet.
- Formulated and optimized complex composite loss landscapes, utilizing automatic differentiation to enforce Surface Energy Balance (SEB) and Stokes flow constraints on model outputs.
- Engineered end-to-end data pipelines for high-resolution climatological data, implementing timestamp transformations and gradual introduction of loss terms to overcome gradient problems in LSTM learning.

Certifications

Data Science for Engineers – NPTEL, IIT Madras (Oct 2022)

Data Science Professional Certificate – IBM via Coursera (Jun 2022)

Machine Learning – Stanford Online via Coursera (Aug 2021)

IELTS Academic – Overall Band Score: 8.0 (CEFR C1) (Oct 2022)

Languages

English: Native Proficiency (IELTS 8.0/C1 level)
Swedish: Advanced (B1/B2 level)
Kannada: Fluent
Malayalam: Fluent
Hindi: Advanced